

Outline

Pubertal Development

- Review literature on pubertal development in boys and girls and links to outcomes in adolescence, early adulthood, and mid-life.
 - Pubertal timing (i.e., the onset of puberty) demonstrates significant individual variability, with far reaching consequences for developmental processes.
 - In particular, early maturing girls have higher risk for mood disorders and externalizing behaviors such as delinquency, substance use, early sexual activity.
 - Both early- and late-maturing boys demonstrate elevated risk for depression and other internalizing behaviors
 - Comparatively, relatively little is known about the long-term impact of pubertal processes beyond adolescence and early adulthood but:
 - Some evidence suggests that links between pubertal timing and adverse outcomes are minimal by the fourth decade of life (Boden, 2011; Senia, 2018)
 - Other evidence suggests that risk conferred by early or off-time pubertal development persists into adulthood in certain domains (e.g., depression, sleep; Hoyt, 2020)
 - Understanding these links between pubertal processes and behavioral outcomes depends on measurement

The Measurement of Pubertal Timing

- What measures are available to measure pubertal timing?
 - Menarche (single indicator), self-report (e.g., PDS, peer relative timing measures), hormone measurements, physical exams
 - Can vary in terms of their objectivity (vs. subjectivity), whether they're concurrent or retrospective, or whether they're cross-sectional or longitudinal
 - Different measures have different strengths
 - Adolescents aren't great at identifying early pubertal signs which can lead to regression through the stages (Tanner Stage 3 → Tanner Stage 2)
 - Some items in the PDS aren't as reliable (e.g., skin changes; Shirtcliff, 2009)
 - Timing might be best assessed after puberty. Explain why.
- Retrospective pubertal measures
 - There are several measures available to assess retrospectively reported pubertal timing including retrospective PDS measures and a timing measure (Beltz & Berenbaum, 2013).
 - Psychometrically validated/assessed → Cronbach's alpha, ICC, correlations with recalled age for menarche & spermarche, but not with prospective measurements or a separate dataset
 - Retrospective timing measures are a powerful tool.
 - Work with these retrospective measurements has revealed important insights into the persistence of mood disorders, substance use, and behavioral problems
 - But concerns about the use of these measures persist:
 - Reliability: high internal consistency at the time of assessment (see Beltz & Berenbaum, 2013), but what about test-retest reliability or reporting over time?

- Validity: what is the role of recall bias → can't look at validity because there's no objective reports of pubertal development during adolescence

What's Known about the Reliability and Validity of Retrospective Reporting?

- Review what sexual initiation and substance use studies tell us about reliability of retrospective reporting as a benchmark. Some key issues include:
 - Length of recall
 - Sex differences
 - Off-time/Regression to the mean
- Review menarche/timing studies (i.e., what gaps does our study fill?)
 - Marie & Jacobsen (2017): High reproducibility of menarche with long length between initial and retrospective report.
 - But both time points were retrospective
 - Gilger et al (1991): substantial test-retest reliability, MZ intraclass correlations exceeded DZ intraclass correlations, magnitude of relations followed theoretical expectations of traits with genetic influences
 - If many retrospective reports are taken at 18, then those might good reports
 - But only measured in adulthood
 - And small sample size
 - General limitations of these studies:
 - Both time points were retrospective
 - Both time points were in adolescence
 - Only menarche → no boys were assessed

The Current Study

- Compare prospective reports of status to a validated measure of retrospective pubertal timing using a large cohort of youth followed prospectively into adulthood
 - Aim 1: Examine the correspondence between prospective and retrospective reports of timing in boys and girls.
 - Compare these findings to sexual initiation
 - Aim 2: Examine whether prospective and retrospective reports of timing are differentially associated with outcomes.
 - Aim 3: Examine factors (moderate the accuracy of outcomes) associated with better or worse recall of outcomes (e.g., age, length of recall...)

Methods

Participants

Measures

Analytic Plan

Results

Aim 1: Examine the correspondence between prospective and retrospective reports of timing in boys and girls.

- Report means and zero order correlations for indicator (sexual initiation, menarche, girls' timing, boys' timing) in text
- Figure 1a & 1b: Average sexual initiation and menarche by early, on-time, and late categories
- Figure 2c & 2d: Average sexual girls' and boys' timing by early, on-time, and late categories
- Figure 3. Correspondence between prospective and retrospective reports of timing.

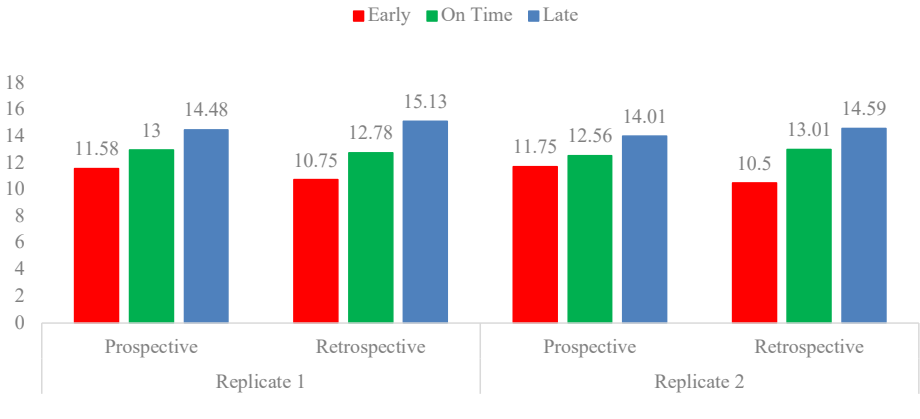


Figure 1b. EXAMPLE menarche by early, on-time, and late categories

Commented [NC1]: need to add error bars from std. deviation and N's

- Early
- On-time
- Late

Commented [NC2]: girls' timing

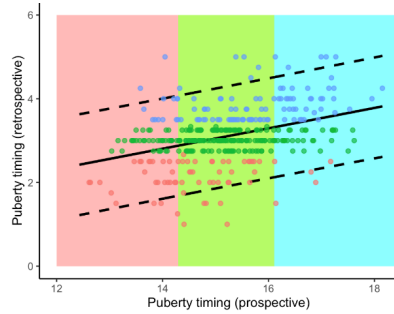
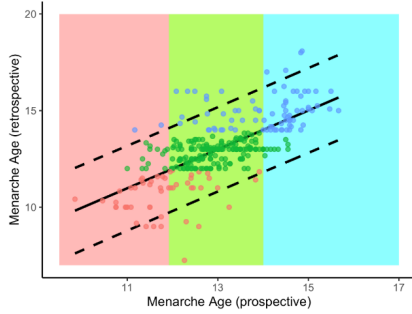
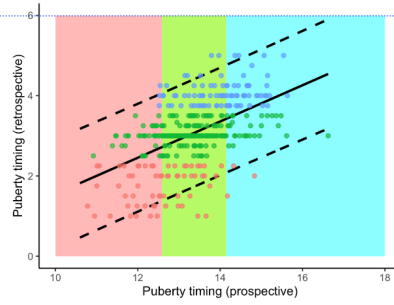


Figure 3. EXAMPLE Correspondence between prospective and retrospective reports of timing.

Commented [NC3]: Need to grab sexual initiation

Aim 2: Examine whether prospective and retrospective reports of timing are differentially associated with outcomes.

- Figure 4. Depict outcomes using area plots (e.g., Beltz, 2014)

Aim 3: Examine factors (moderate the accuracy of outcomes) associated with better or worse recall of outcomes (e.g., age, length of recall...)

- Report in text

Discussion

**TRANSPORTATION POOLED FUND PROGRAM
QUARTERLY PROGRESS REPORT**

Date: April 26, 2016

Lead Agency (FHWA or State DOT): Washington State Department of Transportation

INSTRUCTIONS:

Project Managers and/or research project investigators should complete a quarterly progress report for each calendar quarter during which the projects are active. Please provide a project schedule status of the research activities tied to each task that is defined in the proposal; a percentage completion of each task; a concise discussion (2 or 3 sentences) of the current status, including accomplishments and problems encountered, if any. List all tasks, even if no work was done during this period.

Transportation Pooled Fund Program Project # (i.e., SPR-2(XXX), SPR-3(XXX) or TPF-5(XXX))		Transportation Pooled Fund Program -- Report Period:	
TPF-5(343) Roadside Safety Research for MASH Implementation		☐ Quarter 1 (January 1—March 31) XX	
		☐ Quarter 2 (April 1—June 30)	
		☐ Quarter 3 (July 1—September 30)	
		☐ Quarter 4 (October 1—December 31)	
Project Title:			
Roadside Safety Research Program			
Name of Project Manager(s): Andrew Beagle	Phone Number: 360-705-7978	E-Mail beaglea@wsdot.wa.gov	
Lead Agency Project ID: N/A	Other Project ID (i.e., contract #):	Project Start Date: 2016	
Original Project End Date:	Current Project End Date: 2020	Number of Extensions:	

Project schedule status:

XX ☐ On schedule ☐ On revised schedule ☐ Ahead of schedule ☐ Behind schedule

Overall Project Statistics:

Total Project Budget	Total Cost to Date for Project	Percentage of Work Completed to Date
\$570,000 (currently)	TBD	0%

Quarterly Project Statistics:

Total Project Expenses	Total Amount of Funds	Total Percentage of
------------------------	-----------------------	---------------------

TPF Program Standard Quarterly Reporting Format — 9/2011 (revised)

and Percentage This Quarter	Expended This Quarter	Time Used to Date
0	0	0

Project Description:

The Roadside Safety Research for MASH Implementation program is designed to conduct research on roadside safety priorities for research projects aligned with the MASH implementation completion schedule. The compliance dates for MASH roadside safety hardware are:

- December 31, 2017: W-beam barriers and cast-in-place concrete barriers
- June 30, 2018: W-beam terminals
- December 31, 2018: Cable barriers, cable barrier terminals, crash cushions
- December 31, 2019: Bridge rails, transitions, all other longitudinal barriers (including portable barriers installed permanently), all other terminals, sign supports, and other breakaway hardware.
- Also, temporary work zone devices, including portable barriers, manufactured after December 31, 2019, must have been successfully tested to the 2015 edition of MASH.

Washington State Department of Transportation is the lead agency for this study. Texas A&M Transportation Institute (TTI) will conduct the research for the funded projects. A technical committee made up of a representative from each participating state will manage the program. TTI will report to and work under the guidance of the technical committee. Pooled fund contributions will be used to pay for travel expenses for a state representative to attend the annual pooled fund meeting.

The pooled fund research program will identify, analyze, and develop solutions for roadside safety problems with the goal of reducing the tremendous loss of life that occurs on our highways each year as a result of roadway departure crashes. Specific research activities addressed within the program will include the design, analysis, testing, and evaluation of roadside safety hardware, and the development of guidelines for the use, selection, and placement of these features. Research problem statements will be developed by participating member state representatives. The members will rank and select specific projects to be funded each fiscal year. Additionally, member states may independently develop and fund research projects through the Roadside Safety Pooled Fund Program through a reimbursable agreement with Washington DOT.

The Roadside Safety Committee invites other state DOT's to join this collaborative research effort by making modest contributions to the pooled fund and participating in the development of the research inquiries. States that are interested in joining this collaborative research effort should contact Andrew Beagle at (360) 705-7978.

Progress this Quarter (includes meetings, work plan status, contract status, significant progress, etc.):

Initial meetings have been held between WSDOT and TTI planning the study plan and prepare for the first technical meeting. Master agreement with TTI is in place for work to initiate, but awaiting obligation transfers to begin the first project "MASH Coordination Effort" which will identify a recommended study plan in line with common hardware used by states and to help guide the Roadside Technical Committee for their upcoming project selection meeting in Fall 2017.

Anticipated work next quarter:

Website development for pooled fund (this will adapt the previous website used for TPF-5(114) <http://www.RoadsidePooledFund.org> to this pooled fund).
Begin study "MASH Coordination Effort".

|

--

Significant Results: TBD

Circumstance affecting project or budget. (Please describe any challenges encountered or anticipated that might affect the completion of the project within the time, scope and fiscal constraints set forth in the agreement, along with recommended solutions to those problems). No issues at this time.
--

|

--

Potential Implementation: Expect many research project inquires to result in modified designs which have been accepted by FHWA.
--